

Cell Wall Inhibitors 2

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Cephalosporins

- Cephalosporins also have a **β-lactam ring** and have the same mechanisms of action like penicillin.
- Each newer generation of cephalosporins is **increasingly resistant to β-lactamase.**

- **Antibacterial spectrum And Classification:**
- They are divided into 1st, 2nd, 3rd, 4th and 5th generations.
- The difference among the groups is marked by changes in antibacterial spectrum. In general, *the activity against gram-positive bacteria decreases from first to third generation while activity against gram-negative organisms increases.*

First-generation cephalosporins

Gram (+) cocci

- Staphylococcus aureus*
- Staphylococcus epidermidis
- Streptococcus pneumoniae
- Streptococcus pyogenes
- Anaerobic streptococci

Gram (–) rods

- Escherichia coli
- Klebsiella pneumoniae
- Proteus mirabilis

Second-generation cephalosporins

Gram (+) cocci

- Staphylococcus aureus*
- Streptococcus pneumoniae
- Streptococcus pyogenes
- Anaerobic streptococci

Gram (–) cocci

- Neisseria gonorrhoeae

Gram (–) rods

- Enterobacter aerogenes
- Escherichia coli
- Haemophilus influenzae
- Klebsiella pneumoniae
- Proteus mirabilis

Anaerobic organisms**

Summary of therapeutic applications of cephalosporins. **Methicillin*-resistant staphylococci are resistant. ***Cefoxitin* and *cefotetan* have anaerobic coverage. †*Ceftazidime* only.

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Third-generation cephalosporins

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- Neisseria gonorrhoeae

Gram (–) rods

- Enterobacter aerogenes
- Escherichia coli
- Haemophilus influenzae
- Klebsiella pneumoniae
- Proteus mirabilis
- Pseudomonas aeruginosa†
- Serratia marcescens

Fourth-generation cephalosporins

Antibacterial coverage comparable to that of the third-generation class; however, demonstrate greater stability against β-lactamases

1. First generation cephalosporins : Cephalexin, Cefazolin, cefadroxil

- Primarily cover **gram-positive organisms** (similar to penicillin G) with some **gram-negative coverage (E. coil, Klebsiella)**, making them effective in urinary tract infection.
- They are sensitive to β-lactamase.
- They do not penetrate to CSF (even in presence of meningitis).
- There is partial cross hypersensivity between them and penicillins.

2. Second generation cephalosporins :

Cefoxitin, cefuroxime, cefaclor.

- They maintain **gram-positive coverage** with enhanced coverage of **gram-negative bacteria** e.g. H. influenza, Neisseria, Proteus, E. coli and K. pneumoniae (but not Pseudomonas).

- They are used primarily in the management of: 1. urinary and respiratory tract, 2. bone, and soft-tissue infections and 3. prophylactically in various surgical procedures (abdominal colorectal and gynecologic procedures).

- They are relatively resistant to B-lactamases.
- They do not penetrate to CSF (even in presence of meningitis, except cefuroxime).
- Cefotetan and cefoxitin have appreciable activity against gram-negative anaerobic bacteria (for example, Bacteroides fragilis).

3. Third generation cephalosporins :

Cefotaxime, Ceftazidine, Cefoperazone, Ceftriaxone, Cefdinir, Cefixime

- They maintain gram-positive coverage with excellent coverage of gram-negative bacteria(**including Pseudomonas**)and active against **anaerobes.**
- They can penetrate to CSF in presence of meningitis (except **cefoperazone**).
- They are excreted by the kidney, except **cefoperazone has biliary excretion and ceftriaxone has dual elemiation renal and biliaery** which are excreted biliary allowing their use in renal failure.
- They are highly resistant to B-lactamase.
- They have long duration (12 hrs).
- ***Ceftriaxone*** has longest half-life (6 to 8 hours) so it has longer duration (24 hrs) effective against penicillin-resistant **Neisseria gonorrhoeae**. The drug is **excreted in bile and may be used in patients with renal insufficiency. It has good penetration into bone.**

4. fourth generation cephalosporins

- ***Cefepime and cefpirome*** *has a wide antibacterial spectrum.*
- must be administered parenterally.
- Effective against *streptococci* and *staphylococci* (but only MSSA).
- Effective against aerobic gram negative organisms, such as
- Enterobacter species

  E. coli.

  K. pneumoniae.

  P. mirabilis, and

  S. aerouginosa

5. fifth generation cephalosporins

Ceftaroline 

- a broadspectrum,
- is administered IV as a prodrug,
- *It is the B-lactam with activity against MRSA and penicillin-resistant Streptococcus pneumoniae.*
- *Poor activity against p. aeruginosa*
- has similar gram-negative activity to the third-generation cephalosporin ceftriaxone but *poor activity against p. aeruginosa*.
- *It is indicated for the treatment of complicated skin and skin structure infections and community-acquired pneumonia.*

- **Pharmacokinetics:**
  - Cephalosporins are widely distributed in body fluids; members of the third, fourth and fifth generations can penetrate to CSF (except ***cefoperazone***).
  - adequate therapeutic levels in the CSF, regardless of inflammation, are achieved with ***ceftriaxone*** and ***cefotaxime.***
  - Like penicillin, most cephalosporins are excreted via the kidney and hence the dose must be adjusted in patient with renal insufficiency. The renal excretion is also decreased by probenecid.
  - ***Cefoperazone and ceftriaxone*** (third generation) are excreted in bile and its serum level is not greatly influenced by renal failure.

Therapeutic uses:

- Severe undiagnosed sepsis especially in immunosuppressed patient.
- Treatment of infection of respiratory tract, urinary tract, skin, soft tissue, bones and joints due to susceptible organisms.
- Gram-negative bacterial meningitis may be treated by cefotaxime (third generation) and ceftriaxone that reach the C.N.S.
- Biliary infection: 3rd generation (cefoperazone or ceftriaxone).
- Gonorrhoea due to penicillin-resistant Gonococci. It is treated by single IM injection of ceftriaxone.
- Pseudomonal infection when aminoglycosides are not desirable.

Adverse effects:

- Generally well tolerated.
- Cephalosporins should be avoided or used with caution in individuals with penicillin allergy. The highest rate of allergic cross-sensitivity is between *penicillin* and first-generation cephalosporins is around 3% to 5%.
- Superinfection with **C. difficile (pseudomembranous colitis):** Cephalosporins are the main cause of hospital-acquired C. difficile colitis, a potentially life- threatening infection.

Pseudomembranous colitis

- Antibiotic-associated diarrhea or C. difficile colitis), can occur following antibiotic treatment. When antibiotics are given, most of the resident intestinal bacteria are killed with sparing of some types like Clostridia difficile. The normally harmless C. difficile grow rapidly because of lack of competition with other flora, and produce toxins. These toxins damage the inner wall of the intestines and cause abdominal pain, fever and diarrhea. Severe cases can be life-threatening.
- **Treatment:**
  - – Stop all antibiotics whenever possible.
  - – Oral **metronidazole** 250 mg /6 hr (1st choice).
  - – If metronidazole failed, give oral **vancomycin** solution 125 mg/6 hr

	1ST GENERATION	2ND GENERATION	3RD GENERATION	4TH GENERATION
	Cephalexin Cephradine Cefadroxil	**Cefoxitin Cefuroxime Cefaclor**	**Cefotaxime Cefoperazone Ceftriaxone**	**Cefipime**
Spectrum	Mainly Gram +ve spp. and few Gram –ve bacilli	Maintain Gram +ve coverage and enhanced Gram –ve coverage	Excellent Gram –ve coverage (including *Pseudomonas*)	Wide Gram +ve and Gram –ve coverage including *Pseudomonas*
β-lactamase susceptibility	Yes	Yes	No	No
Penetration to CSF	No even in meningitis	No *(except cefuroxime)*	Yes *(except cefoperazone)*	Yes
Cross allergy with penicillin	High	Moderate	Low	Low
Elimination	Renal	Renal	Renal & biliary	Renal & biliary

Monobactams

- The monobactams, betalactam antibiotics which also disrupt bacterial cell wall synthesis.
- **Aztreonam** which is the only commercially available monobactam,
- has antimicrobial activity directed primarily against gram-negative pathogens, including the Enterobacteriaceae and P. aeruginosa.
- It lacks activity against gram positive organisms and anaerobes.

- Aztreonam is resistant to the action of most β-lactamases. It could be used in severe respiratory, urinary, biliary, GIT and female genital tracts infections.

- It is administered either IV or IM and can accumulate in patients with renal failure. Aztreonam is relatively nontoxic, but it may cause phlebitis, skin rash.

- It shows little cross-reactivity with antibodies induced by other β-lactams. Thus, this drug may offer a **safe alternative** for treating patients who are allergic to other penicillins, cephalosporins, or carbapenems and it could be used as as alternative to aminoglycosides.

Vancomycin

- **Inhibits cell wall synthesis** and enhances cell lysis.
- It is active against **gram positive organisms.**
- *MRSA and methicillin-resistant Staphylococcus epidermidis (MRSE) infections, as well as enterococcal fecalis and streptococcal pneumonia infections.*
- **Therapeutic uses:** used in patients with 1. skin and soft tissue infections, 2. infective endocarditis, 3. and nosocomial pneumonia, 4. it is used orally to treat antibiotic-associated enterocolitis (*C. difficile* colitis) because it is not absorbed orally.
- It is given by slow i.v. infusion.
- Rapid infusion of vancomycin may cause anaphylactoid reactions and **"red man" syndrome** (skin rash and flushing due to histamine release).
- ***adverse events*** include 1. nephrotoxicity, 2. infusion-related reactions (red man syndrome and phlebitis), and 3. ototoxicity.

Daptomycin:

- Its a bactericidal concentration-dependent, therefore dosed IV once daily
- Daptomycin is used for treating infections caused by resistant gram-positive organisms, including **MRSA and vancomycin resistant enterococci (VRE), sreptococcus spp and corynebacterium .**
- Daptomycin is indicated for the treatment of complicated skin and skin structure infections.
- It also indicated **in bacteremia caused by S. aureus**, including those with right-sided infective endocarditis.
- Additionally, daptomycin is inactivated by pulmonary surfactants; thus, **it should never be used in the treatment of pneumonia.**

Fosfomycin:

- Fosfomycin is an inhibitor of the MurA enzyme, UDP-N-acetylglucosamine-enolpyruvyltransferase, that catalyzes the first committed step in peptidoglycan synthesis
- acts as a time-dependent inhibitor
- Fosfomycin is a **bactericidal**
- It is indicated for urinary tract infections caused by **E. coli or E. faecalis** and is considered first-line therapy for **acute cystitis.**
- Fosfomycin is rapidly absorbed after oral administration and distributes well to the kidneys, bladder, and prostate.
- The drug is **not metabolized** and excreted in its active form in the urine and feces. It maintains high concentrations in the urine over several days, allowing for a one-time dose for the treatment of urinary tract infections.
- The most commonly reported adverse effects include diarrhea, nausea, and headache.

Polymixins

- The polymixin antibiotics are **polymixin B (**parenteral, topical preparations). and **colistin** (polymixin E) parenteral, topical preparations.
- They exert their antibacterial action by disrupting the **outer cell membrane**.
- They have selective and rapid bactericidal action on **Gram-negative bacilli**, especially *Pseudomonas* and coliform spp.
- They are **not absorbed** from the GIT.
- Due to their **high toxicity**, they are used only **locally** for eye and skin infections.
- However, with the increase in gram-negative resistance, they are now commonly used as rescue therapy for patients with ***multidrug-resistant infections***. Careful dosing and monitoring of adverse effects are important to maximize the safety and efficacy of these agents.

Bacteriology Report

Urine Culture and Sensitivity

Pus cells /H.P.F.	6-7/HPF
Colony Count	> 100,000 / ml [Pathogenic Bacte

Sensitivity Result: **Pseudomonas aeruginosa**

Sulphamethazone & Trimthoprim (SXT)	Resistant
Ampicillin (AMP)	Resistant
Cefotaxime (CTX)	Resistant
Amoxycillin & Clavulanic Acid (AMC)	Resistant
Ampicillin & sulbactum (SAM)	Resistant
Amoxycillin (AML)	Resistant
Tetracycline (TE)	Resistant
Oxacilin (OX)	Resistant
Ceftriaxone (CRO)	Resistant
Amikacin (AK)	Resistant
Doxycycline (DO)	Resistant
Tigecycline (TGC)	Resistant
Ertapenem (ERT)	Resistant
Gentamicin (CN)	Resistant
Imipenem (IPM)	Resistant
Meropenem (MEM)	Resistant
Nitrofurantion (F)	Resistant
Cefepime(FEP)	Resistant
Ciprofloxacin (CIP)	Resistant
Norfloxacin (NOR)	Resistant
Levofloxacin (LEV)	Resistant
Colistin (CT)	Resistant
Cefoxitin (FOX)	Resistant
Piperacillin (PRL)	Resistant

Reference

- **Lippincott's Illustrated Reviews Pharmacology 6th edition**